

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\ln(x+233) - \ln(233)}{x} \leftarrow 0 \overline{\leftarrow 0}$$

$$\stackrel{1)}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{x+233}}{1} = \frac{1}{233}$$

$$\begin{aligned} \stackrel{2)}{=} \lim_{x \rightarrow 0} \frac{\ln\left(\frac{x+233}{233}\right)}{x} &= \lim_{x \rightarrow 0} \frac{\ln\left(\frac{x}{233} + 1\right)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x}{233}}{x} \\ &= \frac{1}{233} \end{aligned}$$

$$\textcircled{7} \lim_{x \rightarrow 2025} \frac{2025^x - 2025^{2025}}{x - 2025} \leftarrow 0 \overline{\leftarrow 0}$$

$$x = t + 2025$$

$$t = x - 2025$$

$$\lim_{t \rightarrow 0} \frac{2025^{t+2025} - 2025^{2025}}{t}$$

$$\stackrel{(1)}{=} L'_{\text{Hosp}}$$

$$\lim_{t \rightarrow 0} \frac{2025^{t+2025} \cdot \ln 2025}{1}$$

$$= 2025^{2025} \cdot \ln 2025$$

(2)

$$a^x - 1 \sim x \ln a \quad (x \rightarrow 0)$$

$$e^{x \ln a} - 1 \sim x \ln a \quad (x \rightarrow 0)$$

$$\lim_{t \rightarrow 0} \frac{2025^{2025^t} (2025^t - 1)}{t} = 2025^{2025} \ln 2025$$

$$y = \left(\frac{x}{x+1} \right)^4 = \frac{x^4}{(x+1)^4}$$

$$y' = \frac{4x^3(x+1)^4 - x^4 \cdot 4(x+1)^3}{(x+1)^8}$$

$$= \frac{4x^3(x+1)^3 (x+1 - x)}{(x+1)^5}$$

$$= \frac{4x^3}{(x+1)^5}$$

$$(2) y = \left(\frac{x}{x+1}\right)^4$$

$$y' = 4 \cdot \left(\frac{x}{x+1}\right)^3 \cdot \left(\frac{x}{x+1}\right)'$$

$$\begin{aligned} \left(1 - \frac{1}{x+1}\right)' &= -\left(-\frac{1}{(x+1)^2}\right) \\ &= \frac{1}{(x+1)^2} \end{aligned}$$

$$= 4 \cdot \left(\frac{x}{x+1}\right)^3 \cdot \frac{1}{(x+1)^2}$$

$$= \frac{4x^3}{(x+1)^5}$$

$$3. f'(x) = \frac{2+x}{x^2+4} > 0$$

$$x < -2 \quad f'(x) < 0 \Rightarrow f(x) \downarrow$$

$$x > -2 \quad f'(x) > 0 \Rightarrow f(x) \uparrow$$

$$f(-2) = \int_{-2}^0 \frac{2+t}{4+t^2} dt$$

极值点: $x = -2$

极小值: $f(-2) = \int_0^{-2} \frac{2+t}{4+t^2} dt$

4.

$u \quad d f'(x) \quad v = f'(x)$

$\int_0^1 x \underbrace{f''(x) dx}_{\rightarrow dv} \quad dv = f''(x) dx$

$$= \int_0^1 u dv = uv \Big|_0^1 - \int_0^1 v du$$

$$= (x \cdot f'(x)) \Big|_0^1 - \int_0^1 f'(x) dx$$

$$= 1 \cdot f'(1) - 0 \cdot f'(0) - (f(1) - f(0))$$

$$= 4 - (2 - 1) = 3$$

⑤

$$f(x) = \begin{cases} \frac{x^2 - 8x}{x+2} & x > -1 \\ \frac{x^2 - 8x}{1 - (x+1)} & x < -1 \end{cases}$$

$$\begin{array}{r} 11 \\ x^2 - 8x \\ \hline -x \\ \hline 8 - x \end{array}$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - v'u}{v^2}$$

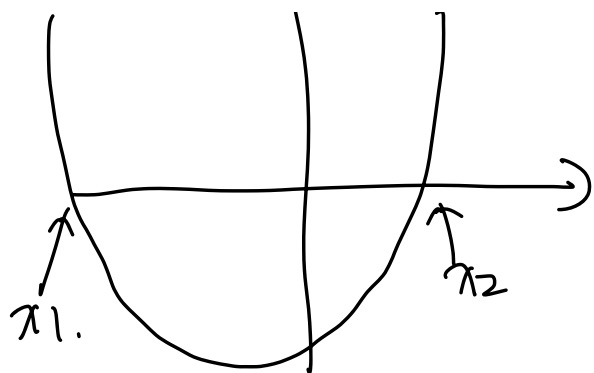
$$f(x) = \begin{cases} \frac{x^2 - 8x}{x+2} & x > -1 \\ 8 - x & x < -1 \end{cases}$$

$$x < -1, f(x) \downarrow$$

$$x > -1, f'(x) = \frac{(2x-8)(x+2) - (x^2-8x)}{(x+2)^2}$$

$$= \frac{x^2 + 4x - 16}{(x+2)^2}$$

$$\uparrow \quad x^2 + 4x - 16$$



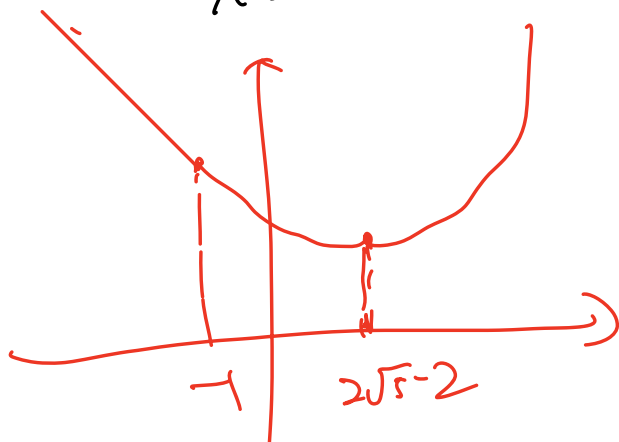
$$x_1 = -2 - 2\sqrt{5}$$

$$x_2 = -2 + 2\sqrt{5}$$

when,

if $x < -2 + 2\sqrt{5}$ then, $f'(x) < 0$ $f(x) \downarrow$

$x > -2 + 2\sqrt{5}$ then $f'(x) > 0$ $f(x) \uparrow$



最小值 $f(2\sqrt{5}-2)$

$$= \frac{(2\sqrt{5}-2)^2 - 8(2\sqrt{5}-2)}{2\sqrt{5}}$$

$$= \frac{40 - 24\sqrt{5}}{2\sqrt{5}}$$

$$= 4\sqrt{5} - 12$$

求 $\frac{x^2 - 8x}{x+2}$ ($x > -1$) 的最小值

令 $t = x+2 > 0$, $x = t-2$

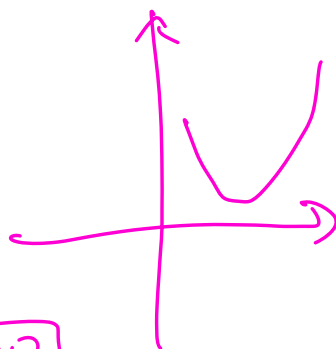
$$\text{上式} = \frac{(t-2)^2 - 8(t-2)}{t}$$

$$= \frac{t^2 - 12t + 20}{t}$$

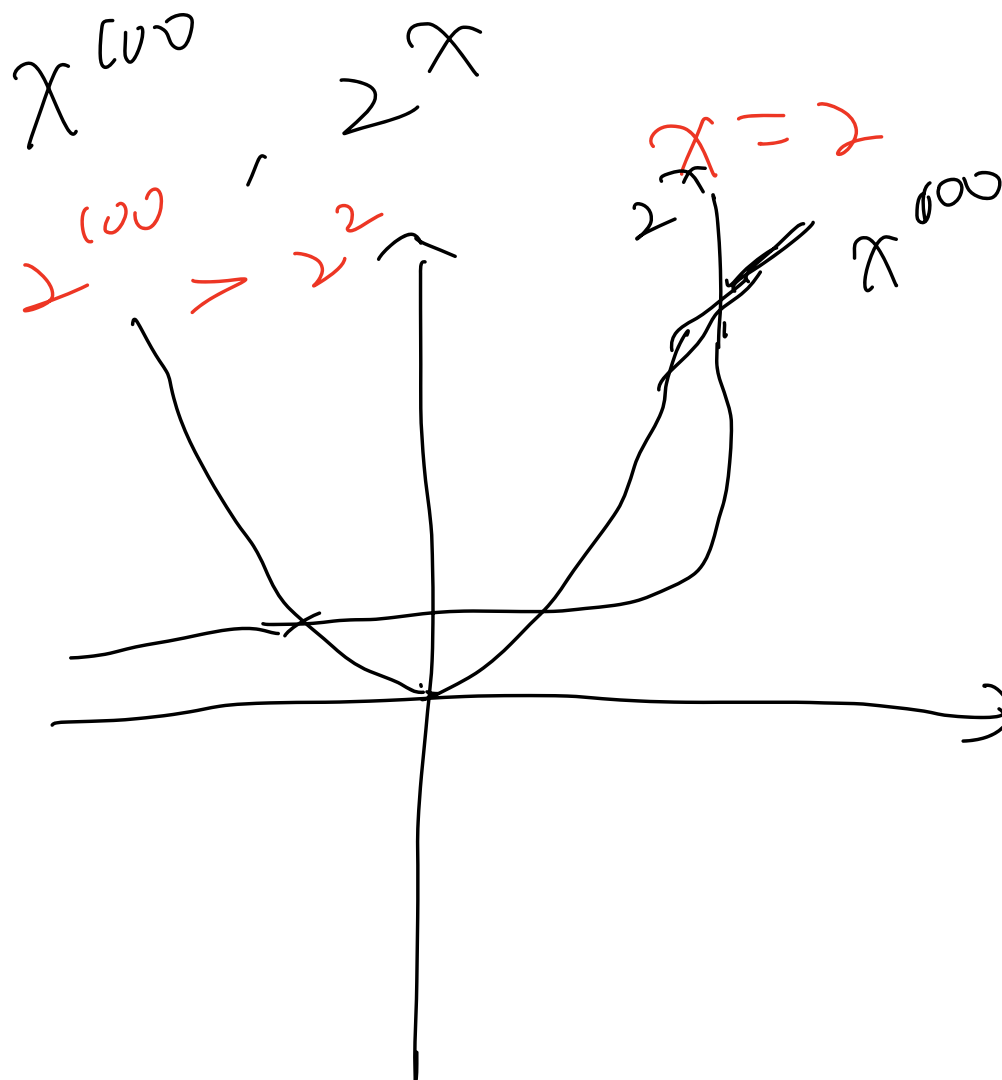
$$= \sqrt{t} - 12 + \sqrt{\frac{20}{t}}$$

$$\geq 2\sqrt{t \cdot \frac{20}{t}} - 12$$

$$= 4\sqrt{5} - 12$$



当 $t = \frac{20}{t}$ 时 ~~求~~ 1) 7.
 $t = 2\sqrt{5}$



$$x > 0$$

$$x^{100} = 2^x$$

$$100 \ln x = x \ln 2$$

$$\frac{\ln x}{x} = \boxed{\frac{\ln 2}{100}} < \underline{0.01}$$

画图像!!!

